

MATLAB/Simulink Sensor Filtering and CAN Telemetry Simulation

Supporting simulation report for STM32 CAN Telemetry Node - Rev A

Overview

This report documents the MATLAB/Simulink telemetry simulation created to support the STM32 CAN Telemetry Node Rev A project. The simulation models four analog sensor channels, converts the signals into STM32-style ADC counts, applies filtering, and demonstrates the planned CAN payload structure for telemetry frame ID 0x100.

Project Status

This is a simulation and documentation artifact. It does not claim physical PCB, ADC, or CAN bus validation. The Rev A PCB and starter firmware are prepared for planned hardware bring-up after board assembly.

CAN Payload Format

CAN ID	DLC	Bytes	Signal	Description
0x100	8	Byte 0-1	AIN1_RAW	Raw ADC value from AIN1
0x100	8	Byte 2-3	AIN2_RAW	Raw ADC value from AIN2
0x100	8	Byte 4-5	AIN3_RAW	Raw ADC value from AIN3
0x100	8	Byte 6-7	AIN4_RAW	Raw ADC value from AIN4

Signal Path

Analog sensor input -> resistor-divider scaling -> STM32 ADC count estimate -> filtering -> CAN ID 0x100 payload formatting.

MATLAB Simulation Outputs

The MATLAB simulation generates synthetic four-channel telemetry data, reconstructs sensor-side voltages from ADC counts, applies moving-average filtering, and packs the last sample into the planned CAN payload format.

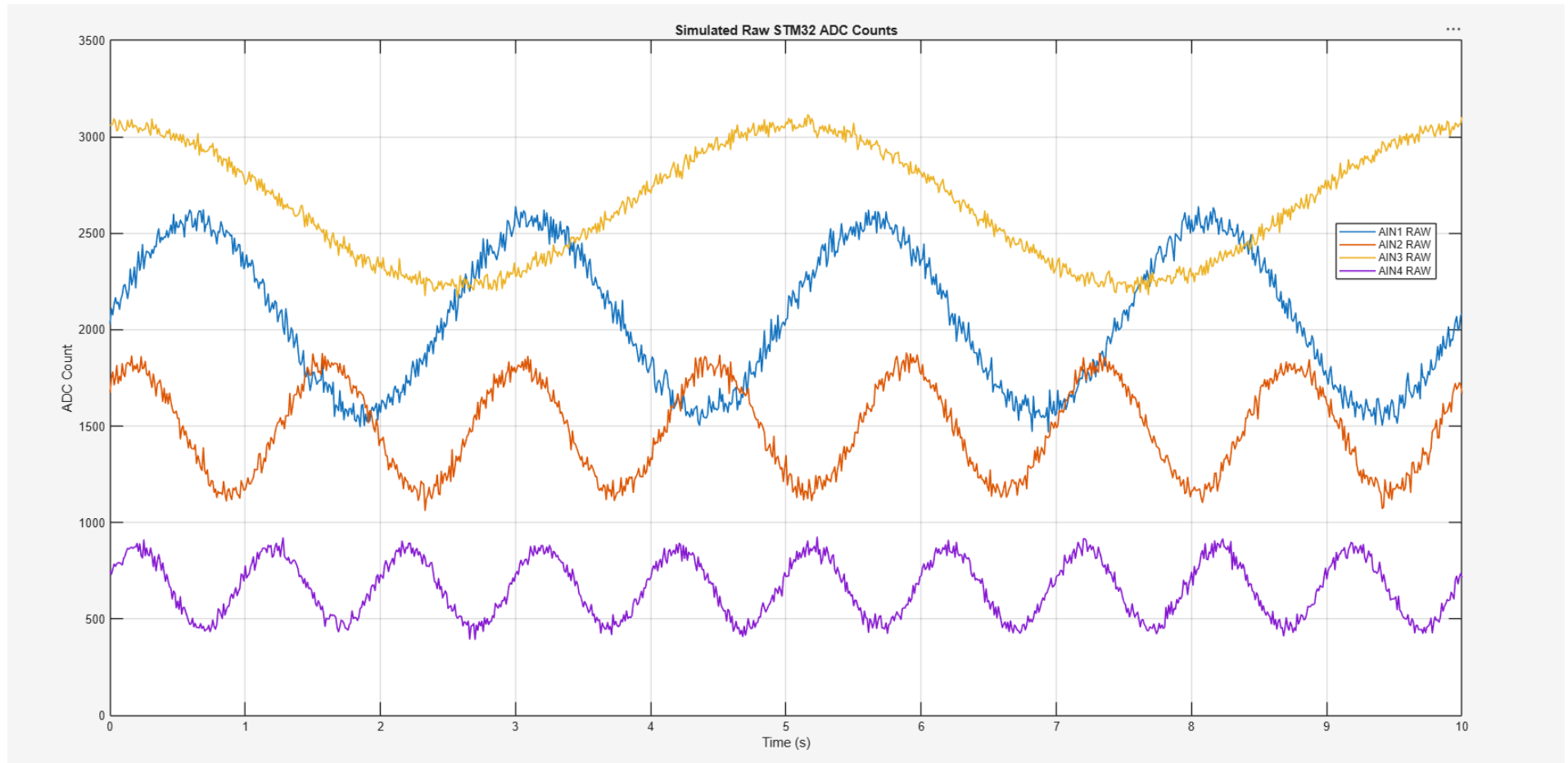


Figure 1. Simulated raw STM32 ADC counts for four analog telemetry channels.

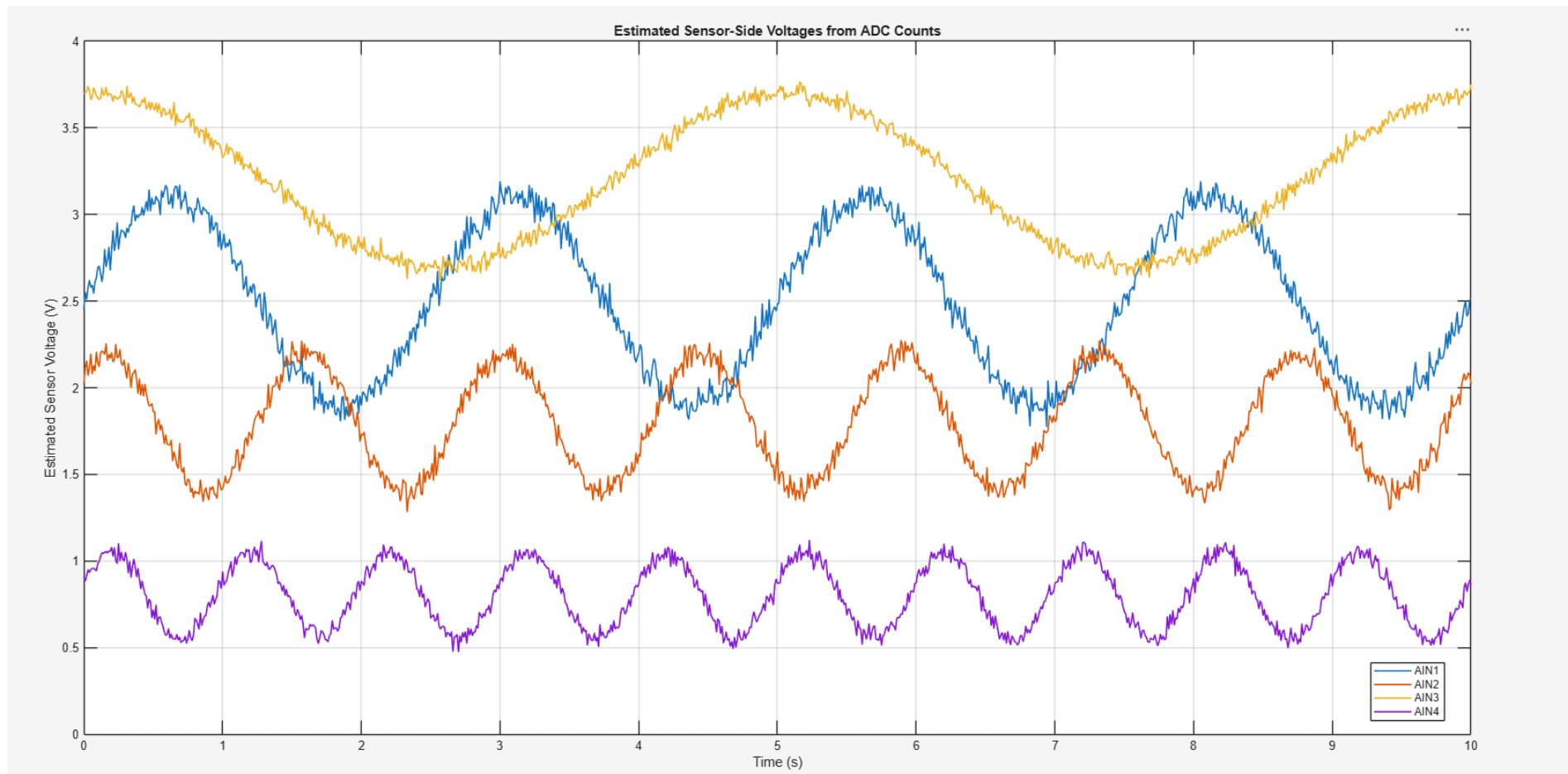


Figure 2. Estimated sensor-side voltages reconstructed from ADC counts using the Rev A resistor-divider model.

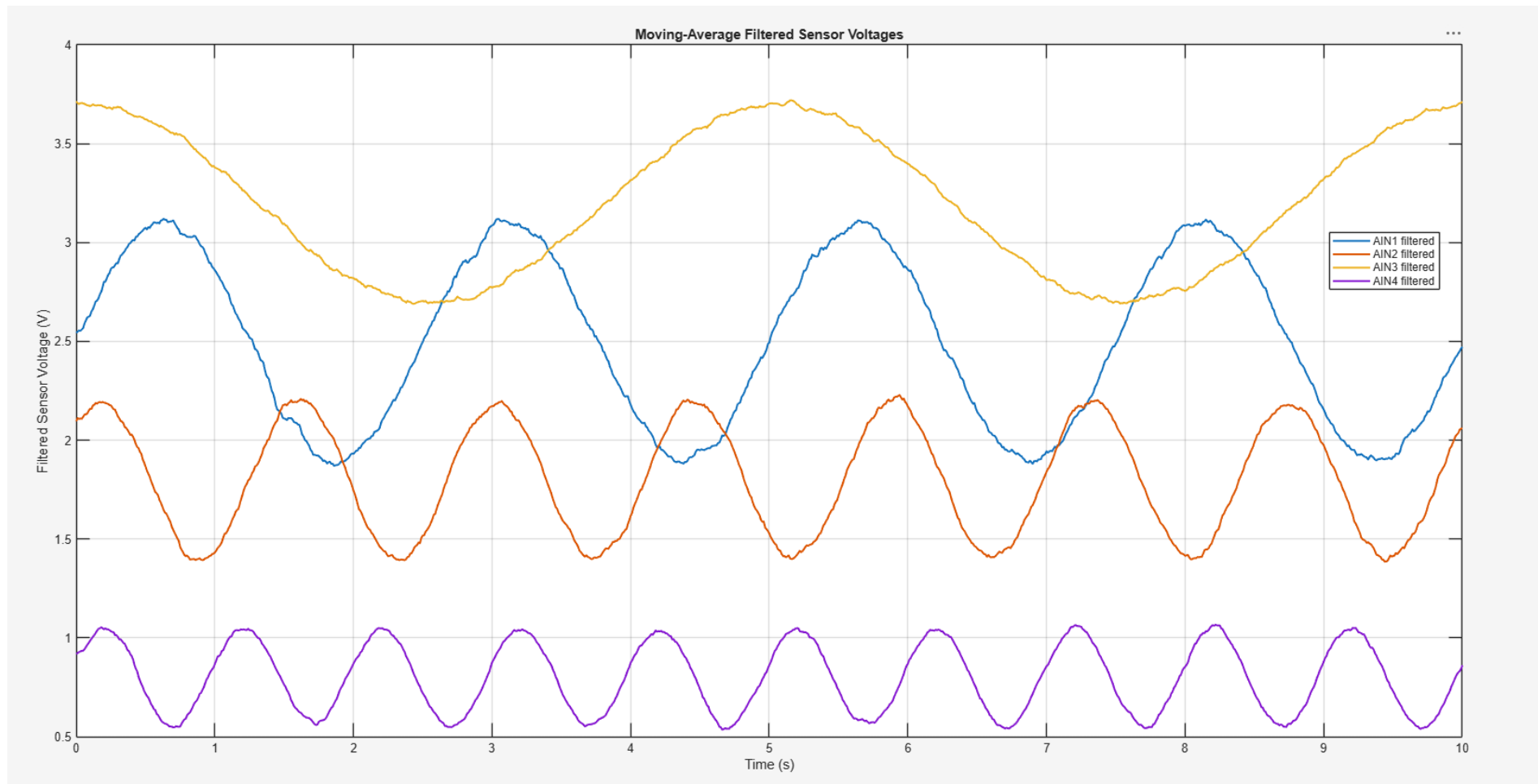


Figure 3. Moving-average filtered sensor voltage estimates for AIN1-AIN4.

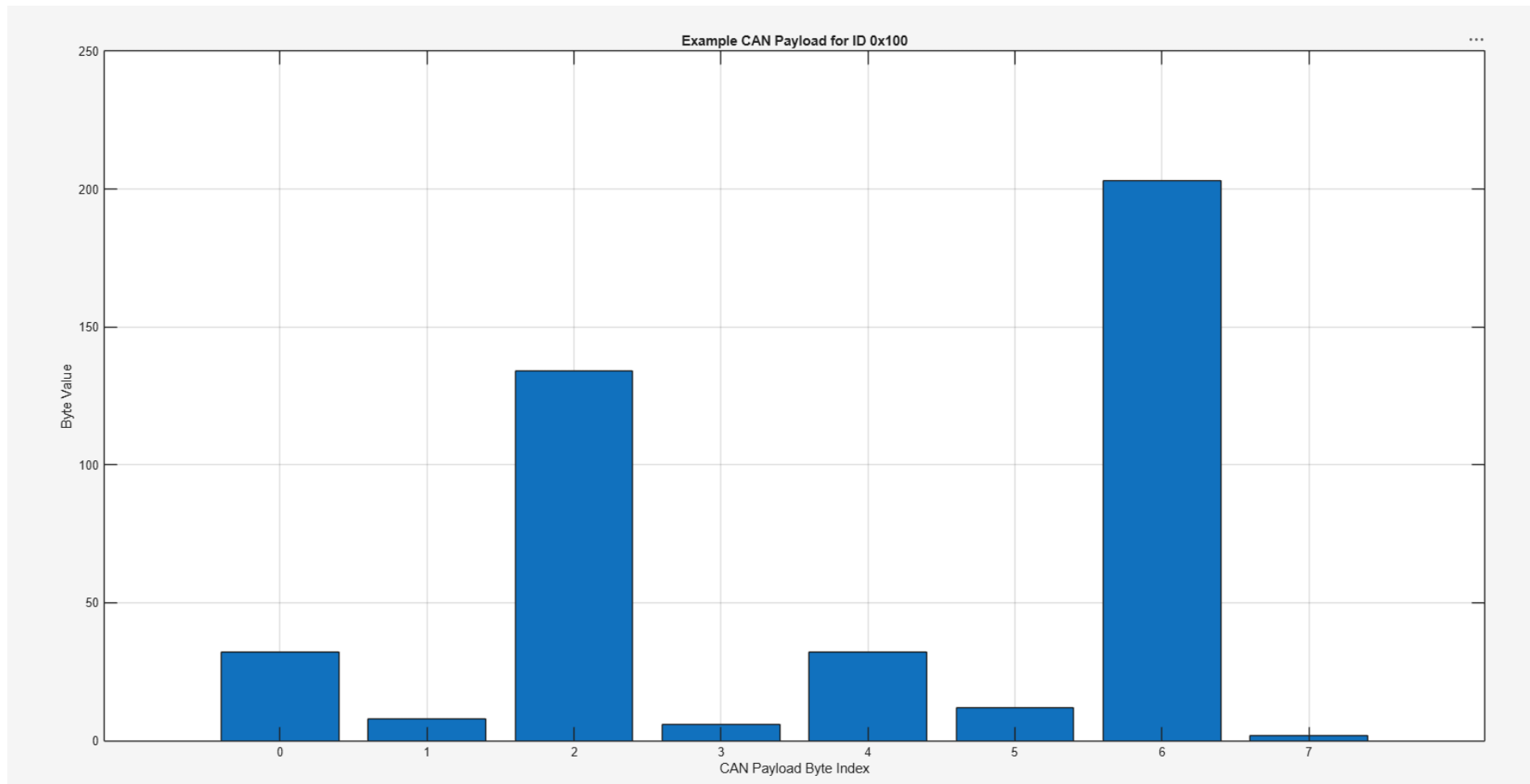


Figure 4. Example 8-byte CAN payload for standard ID 0x100, showing little-endian packing of four raw ADC values.

Simulink Model and Scope Outputs

The Simulink model visualizes the planned telemetry signal path for AIN1-AIN4. Each channel includes a simulated sensor input, Rev A divider scaling, ADC count conversion, and a first-order discrete low-pass filter before the filtered telemetry scope output.

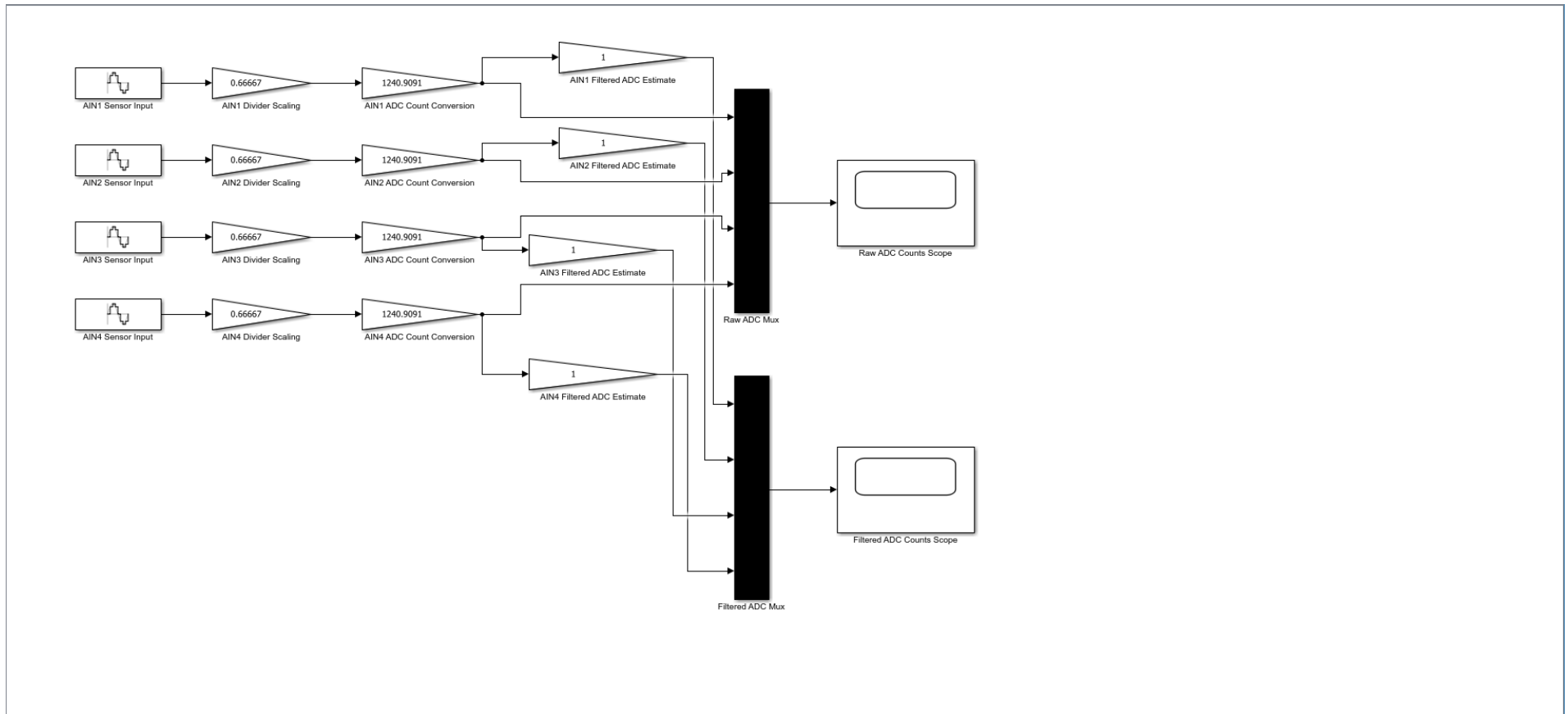


Figure 5. Simulink signal-flow model showing four sensor channels, divider scaling, ADC count conversion, discrete low-pass filtering, mux blocks, and scopes.

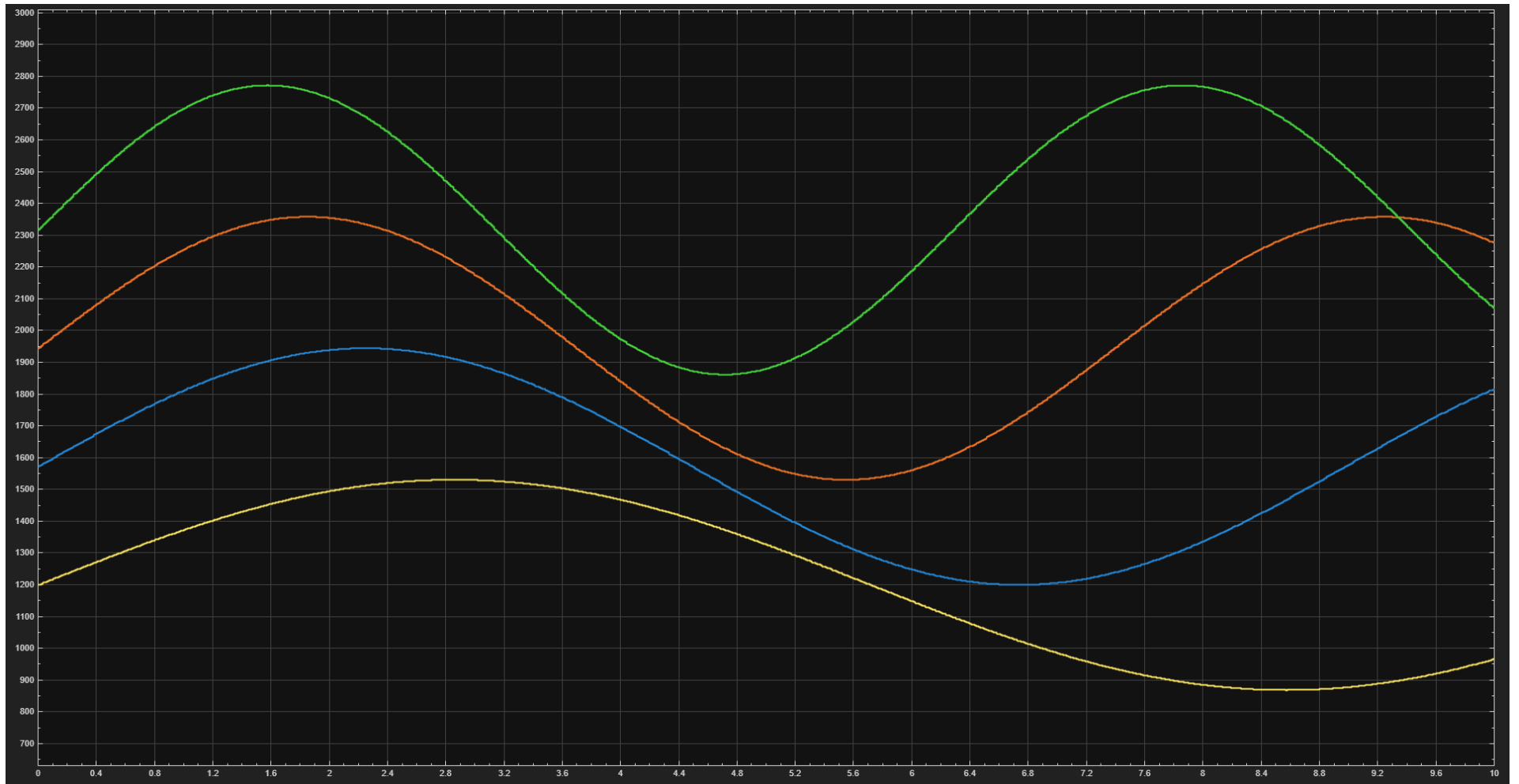


Figure 6. Simulink raw ADC counts scope output for the four simulated telemetry channels.

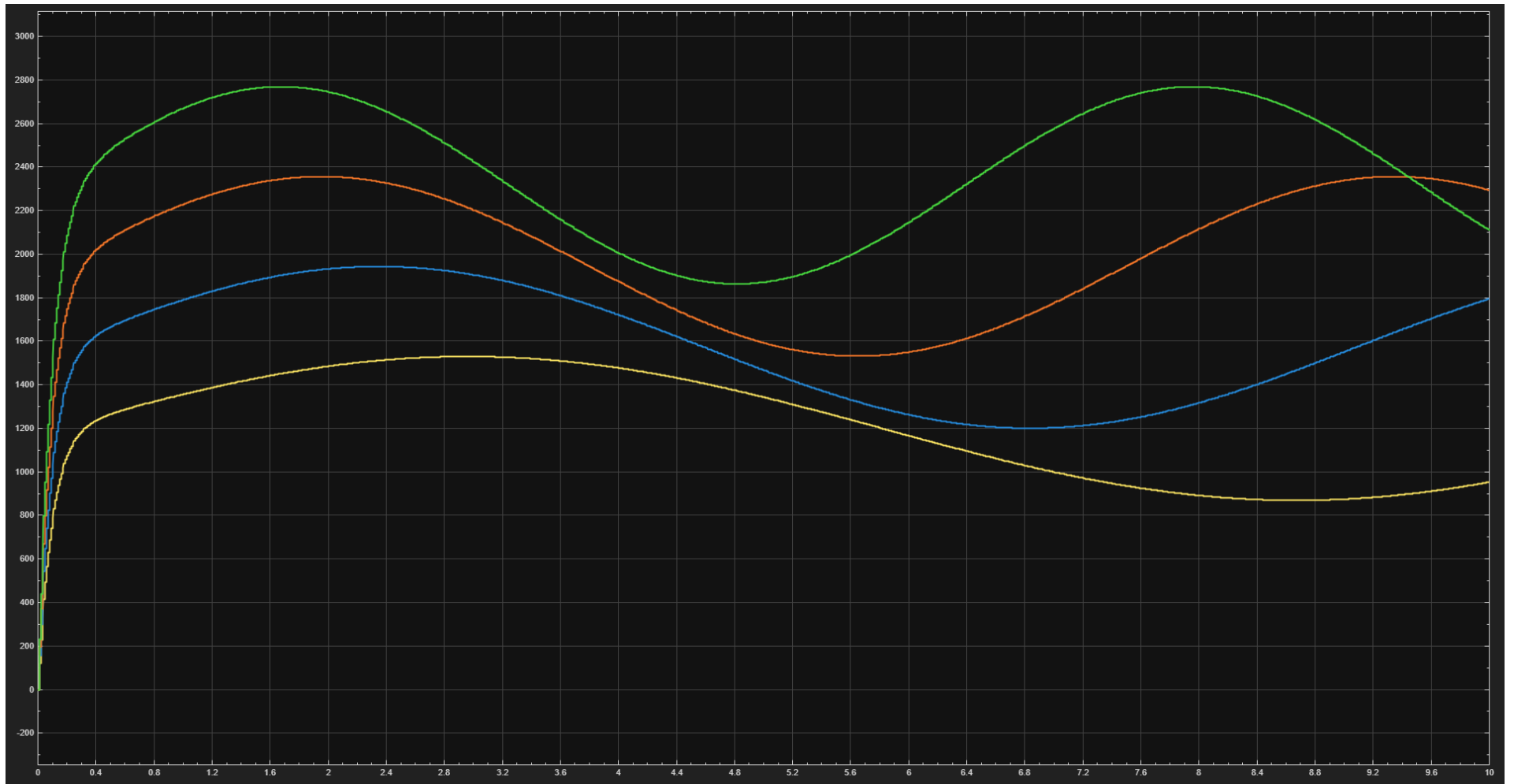


Figure 7. Simulink filtered ADC counts scope output after first-order discrete low-pass filtering.

Implementation Notes

MATLAB script responsibilities:

- Synthetic four-channel sensor generation
- 12-bit ADC conversion using 3.3 V reference
- Rev A resistor-divider model using 10 kOhm / 20 kOhm values
- Moving-average filtering
- Little-endian packing of four 16-bit ADC values into an 8-byte CAN payload

Simulink model responsibilities:

- Visual signal-flow model for AIN1-AIN4
- Divider scaling and ADC count conversion blocks
- First-order discrete low-pass filtering
- Raw and filtered ADC scope visualization

Resume summary:

Developed MATLAB/Simulink telemetry simulations for the STM32 CAN node, modeling ADC scaling, sensor filtering, and CAN ID 0x100 payload formatting before hardware bring-up.